Team 37 : Centralized College Merchandise Management System

PROBLEM

Multiple college clubs manage merchandise independently—separate forms, size charts—causing repeated data entry, size-format confusion, and no single view of offerings.

PROPOSED SOLUTION

A unified web portal where all clubs list/sell merchandise. Students create a one-time profile with saved size preferences to streamline every future order.

HOW USERS INTERACT (Students)

Browse Unified Catalog → Place Orders with Saved Profile → Receive Real-Time Delivery/Slot Notifications

CLUB MANAGERS

Dedicated Dashboard → Publish Merchandise & Manage Listings → Track Student Orders & Reach Wider Audience

TECHNOLOGY & DOMAIN

Technology Stack: MERN (MongoDB, Express.js, React, Node.js) | 3-Tier Architecture.

Domain: E-Commerce / Education Technology—Simplifying Campus Merchandise Management.

Key Requirements

Functional Requirements

- **Unified Merchandise Catalog:** Students browse all club merchandise from a single platform — no more jumping between multiple forms.
- **Saved Size Profile:** Students register once and store preferred size; the profile auto-fills on every subsequent order.
- **Order Placement:** Students select items and place orders seamlessly using their saved profile data.
- **Real-Time Delivery Notifications:** Automatic alerts for upcoming delivery and distribution time slots so students never miss a pickup.
- **Club Manager Dashboard:** Clubs publish merchandise listings, view and manage incoming student orders through a dedicated interface.
- **Role-Based Access Control:** Separate login flows and permissions for students and club administrators.

Non-Functional Requirements

- **Scalability:** Handles growing number of clubs, students, and merchandise listings without performance degradation.
- **Usability:** Clean, intuitive UI with minimal steps — browsing, ordering, and profile setup should feel effortless.
- **Security:** The system must ensure secure authentication and authorization so that only authorized users can access protected resources.
- **Maintainability:** The system should be modular and structured to allow easy updates, debugging, and addition of new features.
- **Reliability:** Notification and order systems must be dependable — missed delivery alerts or lost orders are unacceptable.
- **Performance:** Catalog browsing and order operations should respond within acceptable latency even under concurrent usage.